

EDUCATION

St. Thomas Public School | *SSC : 82%*

FEB 2021

Universal Junior College | *HSC : 74%*

FEB 2023

AISSMS IOIT

AUG 2023 – DEC 2027

Bachelor of Technology in Electronics and Telecommunications | CGPA 8.64

SKILLS

- **Core Electronics:** Analog & Digital Electronics, Signal Conditioning, Power Supply Basics
- **Embedded Systems:** Arduino, ESP8266, PIC18F4550
- **Sensors & Interfacing:** Load Cell (HX711), Ultrasonic Sensor, OLED (I2C)
- **Programming:** C, Embedded C, Python, Linux, Ubuntu
- **Tools & Simulation:** Multisim 11.0, Arduino IDE, PCB Design, Proteus
- **Hardware Skills:** Circuit Debugging, Hardware Testing, Fault Analysis, Documentation

EXPERIENCE

MINILEC INDIA Pvt. Ltd

JUN 2025 – JUL 2025

Technical Intern

Pune, India

- Executed functional testing and validation of industrial electronic protection devices using defined test procedures and safety checklists.
- Performed fault analysis and documented test observations as per safety standards
- Gained exposure to industrial electronics, compliance, and product lifecycle processes

ASTRONIC TECHNOSYSTEM LLP

DEC 2024 – DEC 2024

Intern

Pune, India

- Performed hardware debugging using multimeter and basic lab instruments; identified and resolved connection and component-level issues.
- Assisted in circuit validation, testing procedures, and technical documentation for electronics prototypes.
- Gained hands-on experience in electronics testing, including sensor and module-level verification.

PROJECTS

IoT-Based Weighing Scale Machine | *Embedded C, ESP8266, HX711, Load Cell, Blynk, ThingSpeak*

OCT 2025

- Developed a 300 kg IoT-enabled weighing system using a load cell with HX711 amplifier interfaced to ESP8266
- Implemented real-time weight measurement, calibration, and tare/reset functionality for accurate readings.
- Integrated Blynk for live weight monitoring and ThingSpeak for cloud-based data logging and graphical analysis.
- Designed threshold-based alert logic to detect low-weight and abnormal operating conditions.
- Displayed real-time weight values on an OLED display for local visualization.

Smart Zebra Crossing – Automation & Safety System | *Arduino, Servo Motors, Sensors, Embedded C*

APR 2024

- Built an automated pedestrian and traffic safety system using Arduino-based control logic.
- Implemented servo-controlled barricades synchronized with traffic signal states to manage pedestrian crossings.
- Integrated obstacle detection sensors to prevent unsafe barricade operation.
- Designed signal-based automation and safety interlocks to ensure reliable and fail-safe operation.

Transistor-MOSFET IR Object Detection | *BC547, IRFZ44N MOSFET, IR Sensor, Proteus*

AUG 2023

- Designed a microcontroller-free hardware circuit using BC547 NPN transistor for signal inversion and IRFZ44N MOSFET as a low-side switch to drive LED and buzzer on IR object detection.
- Implemented pull-up resistor logic and active-LOW sensor interfacing; validated design via Proteus simulation before zero PCB hardware build.
- Demonstrated low-cost analog hardware design achieving fast response with no processing delay.

Smart Electric Fence Security System | *ESP32, PIR, GSM (SIM800L), Vibration Sensor*

MAR 2026

- Built an IoT-based intrusion detection system using ESP32 with multi-sensor fusion (PIR + vibration + voltage monitoring) to reduce false positives.
- Integrated SIM800L GSM module for real-time SMS alerts and 95%+ detection rate across lab, campus, and agricultural field testing.
- Designed and validated the complete system through Proteus simulation and real-world hardware testing with solar-powered autonomous operation for remote deployment.

Smart Traffic Signal Simulation | *C, Linux System Programming, IPC, POSIX Threads*

FEB 2026

- Developed a real-time smart traffic signal simulation for a four-way junction using C and Linux system programming.
- Implemented inter-process communication using FIFO pipes and message queues for traffic data handling and congestion alerts.
- Designed multithreaded asynchronous logging using POSIX threads, mutexes, and condition variables for synchronized event management.
- Implemented dynamic traffic-based signal control with signal handling and safe shutdown mechanisms.